

Decoupled Class-II Nelson bound and exact self-coupling residual

TECT verification-first repository · claim A8-CLASSII-DECOUPLED-NELSON-BOUND

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1. Purpose and scope

Scope firewall. This note proves a cutoff-uniform Nelson bound and full-sequence finite-volume measure convergence when the Class-II coefficient field and derivative carrier are independent Gaussian copies. It also proves an exact finite-cutoff Gaussian-divergence identity for the original action. It does *not* prove the self-coupled A7 Gibbs measure. The one remaining load-bearing estimate is an adapted-drift bound; a smart-path interpolation bound is an alternative sufficient route.

1 Pinned model and theorem

Let A be the hash-pinned six-real A1 production quadratic operator on $\mathbb{T}_L^3$, $L = 16$, and let $C = A^{-1}$. Its symbol satisfies

$$A(k) \geq c_{\text{sym}} \langle k \rangle^4 \mathbf{I}_6, \quad c_{\text{sym}} > 0. \quad (1.1)$$

All family masses and the internal-lock matrix are nonnegative. It is therefore sufficient for an explicit lower bound to use

$$c_{\text{sym}} = \inf_{x \geq 0} \frac{Yx^2 + Zx + r}{(1+x)^2}. \quad (1.2)$$

At the production point the minimum occurs at $x_* = (2r - Z)/(2Y - Z)$ and gives

$$x_* = 0.640398746432892, \quad c_{\text{sym}} = 0.108364530587115. \quad (1.3)$$

Let Z and Y be independent copies of the corresponding Gaussian field, coupled across cutoffs through common Fourier modes. The admitted regulators are the common, real-even, uniformly bounded scalar multipliers of A7, with compact support or a uniform Schwartz tail. Put $Z_\Lambda = R_\Lambda Z$ and $Y_\Lambda = R_\Lambda Y$, and $C_\Lambda = R_\Lambda C R_\Lambda^*$.

For $z \in \mathbb{R}^6$, define

$$B(z) = \sum_{A=1}^3 \left[a p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c v_A v_A^T \right], \quad (1.4)$$

where $p_A = 2S_A z$, $v_A = 2(S_A - q_A \mathbf{I})z$, and $q_A = z^T S_A z / (|z|^2 + \varepsilon_\rho)$. The floor is fixed at $\varepsilon_\rho = 10^{-12}$. Write $Y = C^{1/2} \xi_Y$ over one common white noise and put $G_\Lambda = D R_\Lambda C^{1/2}$; M_b denotes multiplication by b . For deterministic $b(x) \geq 0$, set

$$L_b = D^* b D, \quad T_{\Lambda,b} = G_\Lambda^* M_b G_\Lambda, \quad (1.5)$$

and

$$Q_{\Lambda,b}(Y) = \frac{1}{2} (\langle \xi_Y, T_{\Lambda,b} \xi_Y \rangle - \text{Tr } T_{\Lambda,b}). \quad (1.6)$$

The decoupled Class-II composite is $Q_\Lambda(Z, Y) = Q_{\Lambda, B(Z_\Lambda)}(Y)$. The local interaction is

$$U_\Lambda(Z) = \int_{\mathbb{T}_L^3} \left(\frac{\lambda}{4} \rho_\Lambda^2 + \frac{\gamma}{6} \rho_\Lambda^3 \right) dx, \quad \rho_\Lambda = |Z_\Lambda|^2, \quad (1.7)$$

with $\lambda = -0.43$ and $\gamma = 1.62$.

Theorem 1 (decoupled Nelson bound and measure). *For every fixed $p > 0$,*

$$\sup_\Lambda \mathbb{E}_{Z, Y} \exp\{-p[U_\Lambda(Z) + Q_\Lambda(Z, Y)]\} < \infty. \quad (1.8)$$

The actions converge in probability on the common product Gaussian space. For any fixed $p > 1$, the weights converge in L^1 , their partition functions converge to a strictly positive finite limit, and the lifted product-space densities converge in total variation. The projected Galerkin laws converge weakly along the full sequence.

This theorem is about an explicitly declared product-Gaussian model. The physical A7 action is recovered only after identifying $Y = Z$, which is not an innocent substitution under the exponential.

2 Exact spatial-background determinant

Fix a deterministic measurable symmetric matrix field $b \in L^2(\mathbb{T}_L^3; \mathbb{R}^{6 \times 6})$ with $b(x) \geq 0$. At each regulated level, $T_{\Lambda, b}$ is positive trace class (and finite rank for compact-support cutoffs). Diagonalising it and using standard real Gaussian coordinates gives the exact identity

$$\mathbb{E}_Y e^{-pQ_{\Lambda, b}(Y)} = \exp\left\{\frac{1}{2} \text{Tr}(pT_{\Lambda, b})\right\} \det(\mathbf{I} + pT_{\Lambda, b})^{-1/2} = \det {}_2(\mathbf{I} + pT_{\Lambda, b})^{-1/2}. \quad (2.1)$$

For $t \geq 0$,

$$0 \leq pt - \log(1 + pt) \leq \frac{p^2 t^2}{2}. \quad (2.2)$$

Consequently

$$0 \leq \log \mathbb{E}_Y e^{-pQ_{\Lambda, b}} \leq \frac{p^2}{4} \|T_{\Lambda, b}\|_{\mathfrak{S}_2}^2. \quad (2.3)$$

The point of this package is that (2.3) holds for a spatially varying background, not merely the constant field checked in A7.

3 Cutoff-uniform Schatten bound

Use the normalized Fourier basis and write $\widehat{b}(q) = L^{-3} \int b(x) e^{-iq \cdot x} dx$. For a fixed admitted regulator family put $M_R = \sup_{\Lambda, k} |r_\Lambda(k)| < \infty$. From (1.1), the (k, ℓ) block of $T_{\Lambda, b}$ obeys

$$\|T_{\Lambda, b}(k, \ell)\|_F \leq M_R^2 c_{\text{sym}}^{-1} \frac{\|\widehat{b}(k - \ell)\|_F}{\langle k \rangle \langle \ell \rangle}. \quad (3.1)$$

The two covariance roots give the factor M_R^2 in (3.1), hence M_R^4 after squaring. Cauchy–Schwarz in the convolution variable gives

$$\sup_q \sum_k \frac{1}{\langle k \rangle^2 \langle k - q \rangle^2} \leq \sum_k \langle k \rangle^{-4} =: S_L < \infty. \quad (3.2)$$

Parseval now yields

$$\sup_\Lambda \|T_{\Lambda, b}\|_{\mathfrak{S}_2}^2 \leq \frac{M_R^4 S_L}{L^3 c_{\text{sym}}^2} \|b\|_{L^2}^2. \quad (3.3)$$

The code uses a deliberately elementary enclosure. With $\alpha = 2\pi/L$, the shell $\|n\|_\infty = m$ has $24m^2 + 2$ lattice points and $|n| \geq m$, so

$$S_L \leq 1 + \alpha^{-4} \left(24 \sum_{m \geq 1} m^{-2} + 2 \sum_{m \geq 1} m^{-4} \right) = 1 + \alpha^{-4} \left(4\pi^2 + \frac{\pi^4}{45} \right). \quad (3.4)$$

At $L = 16$, this upper bound is 1752.06849501828. It is coarse but rigorous, cutoff independent, and computed from the pinned torus length.

Approximation by smooth matrix fields extends (2.1)–(3.3) to every positive $b \in L^2$. Moreover, if $b_n \rightarrow b$ in L^2 , the corresponding operators converge in $\mathfrak{S}_2$. This is the continuity used below.

4 Production coefficient and sextic absorption

The production matrix

$$Q_{\Pi} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

has eigenvalues 0.00125901150092606 and 0.00558473849907223, hence $B(z) \geq 0$. Since $|q_A| \leq 1$ and the embedded Pauli matrices have operator norm one,

$$|p_A| \leq 2|z|, \quad |v_A| \leq 4|z|. \quad (4.1)$$

The Frobenius norm of each rank-one term and the triangle inequality give

$$\|B(z)\|_F \leq (12a + 48|b| + 48c)|z|^2 =: \beta_B |z|^2, \quad (4.2)$$

where

$$\beta_B = 0.256499999999936. \quad (4.3)$$

Combining (2.3), (3.3), and (4.2), conditionally on Z , gives

$$\log \mathbb{E}_Y e^{-pQ_\Lambda(Z,Y)} \leq p^2 K_B \int \rho_\Lambda^2 dx, \quad K_B(M_R) = \frac{\beta_B^2 M_R^4 S_L}{4L^3 c_{\text{sym}}^2}. \quad (4.4)$$

The executable contractive subfamily has $M_R = 1$ and the explicit coarse enclosure $K_B(1) = 0.599144492990436$. Every other fixed admitted family has the same theorem with its finite M_R inserted.

Put

$$A_p = p^2 K_B(M_R) + \frac{p \max(-\lambda, 0)}{4}, \quad b_p = \frac{p\gamma}{6}. \quad (4.5)$$

For every $r \geq 0$,

$$A_p r^2 - b_p r^3 \leq \frac{4A_p^3}{27b_p^2}, \quad (4.6)$$

with equality at $r = 2A_p/(3b_p)$. Therefore

$$\mathbb{E}_{Z,Y} e^{-p(U_\Lambda + Q_\Lambda)} \leq \exp \left\{ L^3 \frac{4A_p^3}{27b_p^2} \right\}, \quad (4.7)$$

uniformly in the cutoff. This proves (1.8). The numerical value of the right-hand side is intentionally not advertised as a sharp physical bound; only its finite, cutoff-independent character is load bearing.

5 Full-sequence cutoff removal

The trace-class covariance gives $Z_\Lambda \rightarrow Z$ in $L^6(\Omega \times \mathbb{T}_L^3)$. At fixed positive floor, B is smooth, has quadratic growth, and satisfies $\|DB(z)\| \leq C_{\varepsilon_\rho}(1 + |z|)$, hence

$$\|B(z) - B(w)\|_F \leq C_{\varepsilon_\rho}(1 + |z| + |w|)|z - w|.$$

Gaussian moment bounds therefore imply

$$B(Z_\Lambda) \longrightarrow B(Z) \quad \text{in } L^2(\Omega; L_x^2). \quad (5.1)$$

The preceding $L_x^2 \rightarrow \mathfrak{S}_2$ estimate, its polarized form, and dominated convergence of the common multipliers give

$$T_{\Lambda, B(Z_\Lambda)} \longrightarrow T_{B(Z)} \quad \text{in } L^2(\Omega; \mathfrak{S}_2). \quad (5.2)$$

Here $T_{B(Z)} = (DC^{1/2})^* M_{B(Z)} DC^{1/2}$. Conditionally on Z , the centered quadratic form is a second Gaussian chaos in the independent carrier Y . Its exact isometry is

$$\mathbb{E}_Y |Q_\Lambda - Q|^2 = \frac{1}{2} \|T_{\Lambda, B(Z_\Lambda)} - T_{B(Z)}\|_{\mathfrak{S}_2}^2. \quad (5.3)$$

Thus (5.2) gives L^2 convergence of Q_Λ . The same all-moment Gaussian convergence and the polynomial argument used in A4 give $U_\Lambda \rightarrow U$ in L^1 . Hence the complete actions converge in probability.

Choose any $p > 1$. Equation (4.7) makes the weights uniformly bounded in L^p , so Vitali's theorem gives L^1 convergence. Also $\mathbb{E}Q_\Lambda = 0$ and $\sup_\Lambda \mathbb{E}U_\Lambda < \infty$; Jensen therefore gives a cutoff-uniform positive lower bound on the partition functions. Normalization preserves L^1 convergence, which is total variation convergence on the common product Gaussian space. Combining that density convergence with $R_\Lambda(Z, Y) \rightarrow (Z, Y)$ in probability gives weak convergence of the projected Galerkin laws along the full sequence.

6 Exact finite-cutoff divergence identity

There is a second result which applies to the original self-coupled A7 composite but does not yet exponentiate it. Whiten a finite cutoff as $X_\Lambda = C_\Lambda^{1/2}\xi$ and define

$$b_\Lambda(\xi) = C_\Lambda^{1/2} D^*[B(X_\Lambda)DX_\Lambda]. \quad (6.1)$$

For the standard Gaussian divergence $\delta_\gamma b = \langle \xi, b \rangle - \text{div}_\xi b$,

$$2V_\Lambda^{\text{ren}} = \delta_\gamma b_\Lambda. \quad (6.2)$$

Indeed, the first term is $\langle DX_\Lambda, B(X_\Lambda)DX_\Lambda \rangle$. The derivative of (6.1) has a B term and a DB term. The trace of the first is

$$\text{Tr}(C_\Lambda D^* B(X_\Lambda) D) = \sum_i \int \text{Tr}[B(X_\Lambda) \Gamma_{\Lambda, i}] dx, \quad (6.3)$$

which is twice the exact A7 counterterm. The DB trace contains the same-point kernel

$$D_x C_\Lambda(x, y)|_{y=x} = 0, \quad (6.4)$$

and hence vanishes for every admitted common-even regulator. This is an algebraic reason the exact covariance and evenness restrictions are load-bearing. An asymmetric regulator would require a different identity.

The Cameron–Martin space is

$$\mathcal{H}_A = \text{Dom } A^{1/2} \simeq H^2(\mathbb{T}_L^3; \mathbb{R}^6), \quad \|h\|_{\mathcal{H}_A}^2 = \langle h, Ah \rangle. \quad (6.5)$$

Equations (6.2)–(6.5) identify the correct Boue–Dupuis geometry for the remaining problem.

7 Why the self-coupled gate is still open

Conditioning on $B(X)$ in the original action does not leave the derivative carrier Gaussian: it is the same field. Thus (2.1) cannot simply be reused. The independent and self-coupled distinction is executable even in the scalar finite Gaussian control

$$Q^{\text{dec}}(z, y) = \frac{g}{2} z^2 (y^2 - 1), \quad Q^{\text{self}}(z) = \frac{g}{2} z^2 (z^2 - 1). \quad (7.1)$$

At the pinned control inputs their negative-exponential moments differ by 1.10×10^{-2} . This is a falsifier for silently identifying the two models, not evidence against the desired self-coupled theorem.

For a covariance flow $C_\Lambda = \int J_t J_t^* dt$, write

$$X_\Lambda = \int J_t dW_t, \quad h_\Lambda(u) = \int J_t u_t dt. \quad (7.2)$$

One sufficient load-bearing estimate is: for every $\epsilon > 0$, uniformly in the cutoff and in every progressive drift u ,

$$\mathbb{E} V_\Lambda^{\text{ren}}(X_\Lambda + h_\Lambda(u)) \geq -\epsilon \mathbb{E} \left[\int \|u_t\|_2^2 dt + \|h_\Lambda(u)\|_6^6 \right] - C_\epsilon. \quad (7.3)$$

The local potential supplies a positive multiple of the L^6 term, while the Boue–Dupuis control cost supplies the first term. Choosing ϵ small would prove the desired A7 Nelson bound.

An alternative sufficient smart-path formulation uses independent X, Y and $Z_s = sX + \sqrt{1-s^2}Y$. If $\Phi_\Lambda(s)$ is the logarithm of the interpolated partition function, it is enough to prove

$$\sup_\Lambda \int_0^1 (\Phi'_\Lambda(s))_+ ds < \infty. \quad (7.4)$$

At $s = 0$ this note applies; at $s = 1$ one reaches A7. Differentiation produces DB and D^2B mixed value–derivative connections under the tilted law. A7’s two-point L^2 identity does not sum the arbitrary vertex count created by the exponential.

The designated proof route is dyadic scale freezing: freeze B two shells below the new derivative shell, apply (2.3) conditionally, and prove that the sum of the resulting self-coupling commutators obeys (7.3). No running mass, orientation term, or vacuum counterterm is silently assumed. If that audit produces one, it must be registered as a new renormalization condition.

8 Devil’s-advocate review

1. **“The old homogeneous $-\Theta(\Lambda^{3/2})$ trial disproves the Nelson bound.” DISMISSED.** It disproves a uniform pathwise lower bound. Gaussian integration cancels the linear trace through $\det_2$; the remaining deterministic-background cost is quartic and the sextic absorbs it. This does not settle self-coupling.
2. **“A7’s constant frozen determinant already proves this note.” DISMISSED.** The Schatten proof here covers every deterministic spatial L^2 matrix background and supplies the coefficient-field and full-sequence arguments.
3. **“Condition on $B(X)$ and apply the same determinant.” UPHELD as invalid.** The derivative carrier is then not an independent production Gaussian. The scalar negative control detects the substitution.
4. **“The coarse lattice constant is a fitted cutoff number.” DISMISSED.** It follows from the exact sup-norm shell count and zeta sums. Partial finite sums are checks, not the proof.

5. **“The rational floor may hide unbounded coefficient growth.” VALID with mitigation.**
The proof fixes a positive floor, uses $|q_A| \leq 1$, and obtains the global quadratic bound (4.2). Floor removal is excluded.
6. **“The decoupled measure is the physical three-component measure.” UPHELD as false.**
It is a product-space control theorem. The physical claim remains behind (7.3) or (7.4).
7. **“No additional interacting counterterm is now proved.” UPHELD.** Gaussian power counting is favorable, but arbitrary vertex number is not yet controlled. Any surviving local term found by the commutator audit must be recorded explicitly.

9 Reproduction

The fail-closed one-command verifier is

```
python codes/foundations/a8_classii_decoupled_nelson_verify.py
```

It reruns a primary spatial-background operator audit and a non-importing complex-row/determinant audit in a temporary directory, including a full- B finite-dimensional Gaussian-divergence Jacobian check and a parity-breaking negative control. It checks every pinned authority hash before execution and then performs the cross-route contract.

10 References

1. M. Boue and P. Dupuis, “A variational representation for certain functionals of Brownian motion,” *Annals of Probability* 26 (1998), <https://doi.org/10.1214/aop/1022855876>.
2. Y. Hariya, “A variational representation and Prékopa’s theorem for unbounded Wiener functionals,” <https://arxiv.org/abs/1505.02479>.
3. N. Barashkov and M. Gubinelli, “A variational method for Φ_3^4 ,” <https://arxiv.org/abs/1805.10814>.

Result footer

Result ID:	A8-CLASSII-DECOUPLED-NELSON-BOUND
Precise statement:	Arbitrary deterministic spatial PSD backgrounds obey a cutoff-uniform $\det_2/\text{Schatten}$ bound; an independent coefficient/derivative product model has all fixed-p Nelson moments and a full-sequence finite-volume limit.
Scope:	$L=16$; $A1 \ q^{-4}$ covariance; three complex fields; six-real convention; fixed floor; common-even scalar regulators; independent coefficient Z and carrier Y .
Evidence:	Analytic determinant, Fourier trace-ideal, sextic absorption, convergence proof; dual executable audit.
Tier before / after:	new / T5 CLOSED@INDEPENDENT-DERIVATIVE-PRODUCT-GAUSSIAN.
No-overclaim statement:	No self-coupled A7 Gibbs measure, adapted-drift bound, extra-counterterm closure, floor removal, infinite volume, phase transition, BCC, T6, or T7.
Next required action:	Prove A7-CLASSII-SELF-COUPLING-INTERPOLATION via the dyadic frozen-background commutator estimate.